MATS 2026 Work Test

Maryam Berijanian

3/15/2026

1 Non-ergodic Mess3 mixture baseline

For Part 1, I constructed he requested dataset. The three components were

$$C0 : (\alpha = 0.60, x = 0.15), \qquad C1 : (\alpha = 0.79, x = 0.11), \qquad C2 : (\alpha = 0.60, x = 0.49),$$

with equal mixture weights $1/3, 1/3, 1/3$. I chose these parameter similar to Table 2 in the given paper.

The emitted token alphabet was $\{0, 1, 2\}$, with a separate BOS token 3, giving vocabulary size 4. Each raw sampled sequence had length 16, and the model input length was therefore 17 after prepending BOS.

Each individual sequence is generated by a single ergodic Mess3 component, but the overall training distribution is a mixture over distinct components with different parameters. Since the component identity is fixed within a sequence but varies across sequences, the full dataset is a non-ergodic mixture distribution rather than a single ergodic source.

Figure 1 shows the realized component counts in the training split. The counts are close to uniform, as expected from the equal mixture weights. This confirms that the dataset contains a balanced mixture over components.

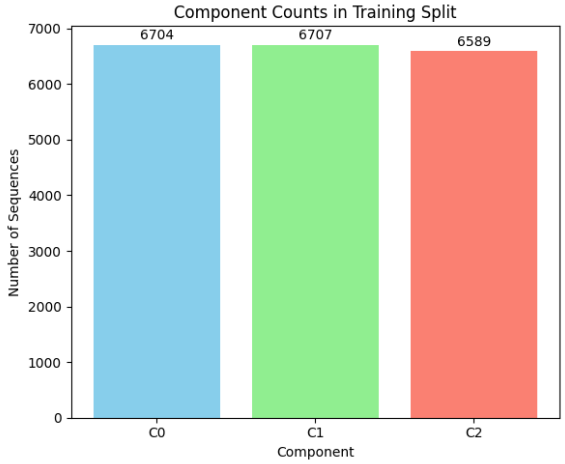

Figure 1: Component counts in the training split.

Table 1 shows one example token sequence from each component. Even though all three components emit tokens from the same alphabet $\{0, 1, 2\}$, they define different sequence distributions, so the transformer must use context to infer which latent regime generated the current sequence.

I then trained a small decoder-only transformer on next-token prediction. The model had 2 layers, model dimension 64, 4 attention heads, and MLP dimension 256. Training used Adam with learning rate 5e-4, batch size 64, and cross-entropy loss. The dataset contained 20,000 training sequences and 2,000 validation sequences.

Figure 2 shows the training and val losses over 10 epochs. Both losses decrease and the total improvement is modest, but it is consistent: the model does learn predictive structure from the dataset. The training and val curves also remain close, which suggests that this small model is learning some useful regularities without obvious overfitting.

Component	Example tokens
C0	0 1 1 2 1 1 2 2 1 1 1 0 0 0 2 1
C1	0 0 0 0 2 2 0 0 0 1 0 1 1 1 0 2
C2	2 1 0 2 1 2 0 1 0 2 2 2 1 1 1 0

Table 1: Example sampled sequences from the Mess3 components used in the dataset.

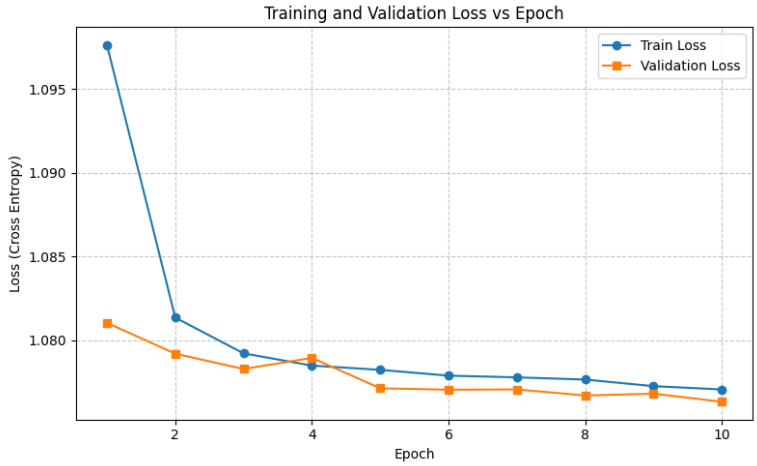

Figure 2: Training and val cross-entropy over 10 epochs on the non-ergodic Mess3 mixture.

This structure is relevant to LMs because natural text is plausibly non-ergodic: different documents or spans may be generated under different latent regimes such as topic, speaker, style, domain, or task. A next-token model trained on such data must infer from context which regime it is currently in, while also tracking within-regime predictive state. Studying non-ergodic mixtures therefore probes whether transformers represent latent regime identity, whether they maintain regime-specific predictive geometries, and whether they organize these uncertainties in a factored or hierarchical way.

2 Pre-registered prediction for activation geometry

According to the paper, predictive vector is the latent state of belief induced by a context, which is sufficient for forecasting the future. The geometry of these predictive vectors should appear in transformer activations.

In our dataset, each sequence is generated by one latent Mess3 component, but the component identity is not observed by the model. Therefore, next-token prediction should depend on whatever latent information in the past is relevant for the future. In my non-ergodic mixture setting, this suggests two relevant latent variables: (1) which Mess3 component generated the current sequence, and (2) conditional on that component, what the current hidden-state belief is within that component. So the activations should organize around uncertainty over component identity as well as uncertainty over within-component Mess3 state.

According to the paper, the predictive vector for a context $x_{1:t}$ is

$$\eta(x_{1:t}) = \frac{\eta^{(0)} T(x_1) \cdots T(x_t)}{\eta^{(0)} T(x_1) \cdots T(x_t) \mathbf{1}},$$

where $\eta^{(0)}$ is the initial latent state and $T(x)$ is "the operator describing latent dynamics for each token x". The important point is that $\eta(x_{1:t})$ is the minimal latent object needed to predict the future. if two contexts give the same predictive vector, they make the same predictions about future tokens.

In my setting, there are three Mess3 components, and each Mess3 process has three hidden states. So the most literal latent state is the pair (C, S_t), where $C \in \{0, 1, 2\}$ is the component identity and $S_t \in \{1, 2, 3\}$ is the current hidden state within that component. This means the full latent state space has $3 \times 3 = 9$ possible

states. Therefore the most literal predictive vector is a normalized belief over 9 joint latent states:

$$\eta(x_{1:t}) = p(C, S_t \mid x_{1:t}).$$

Because this is a normalized distribution over 9 states, it is in an 8-dimensional simplex (because probabilities must sum to 1). So the prediction is:

$$\text{full predictive geometry} \approx \Delta^8.$$

So if the transformer represents the full predictive state in the most straightforward way, I expect the activations to occupy roughly an **8-dimensional predictive state geometry**.

This 8-dimensional count can also be written by the chain rule:

$$p(C, S_t \mid x_{1:t}) = p(C \mid x_{1:t})\, p(S_t \mid C, x_{1:t}).$$

So the full predictive state can be decomposed into (1) a posterior over components $p(C \mid x_{1:t})$, and (2) a within-component hidden-state belief $p(S_t \mid C = c, x_{1:t})$, for each component c.

Since there are 3 components, the posterior $p(C \mid x_{1:t})$ is a probability distribution over 3 values, so it has $3 - 1 = 2$ degrees of freedom. Also, each Mess3 process has 3 hidden states, so each conditional belief $p(S_t \mid C = c, x_{1:t})$ is a probability distribution over 3 values so it also has 2 degrees of freedom. There are three such component-conditional beliefs, so this gives $2 + 3 \times 2 =$ **8 degrees of freedom** total.

So there are two equivalent ways to understand the same formal prediction: as a single predictive vector over 9 joint latent states, which is in Δ^8 and as a decomposition into a 2D posterior over components plus three 2D within-component belief states.

Why this differs from the paper's factorization setting. My setting is related to the paper's predictive-vector framework, but it is not the same as the paper's main factorization setup. In the paper, multiple latent factors run in parallel, the joint state is a tensor product, and under conditional independence there is a lossless lower-dimensional factored representation in orthogonal subspaces.

Here, I do not have three parts running simultaneously. Instead, each sequence comes from exactly one of three possible Mess3 components. So this is a mixture-of-regimes setup rather than a product-of-parts setup. The formal latent state is therefore the pair (C, S_t), where C is component identity and S_t is the hidden state within that component. This gives 9 joint latent states and therefore an 8-dimensional predictive simplex.

Compressed alternative

Although the full predictive state has 8 dimensions, I don't think the transformer necessarily has to realize that geometry in the most explicit way.

Because only one component actually generated each sequence, the posterior over components may become more certain after only a few tokens. If that happens, the model may behave more like it is representing one 2D subspace for uncertainty over component identity, and one shared 2D subspace for the current active Mess3 belief state. This would give an effective geometry closer to $2 + 2 =$ **4 dimensions**.

This 4D picture would not be a full explicit representation of the entire 9-state predictive simplex at all times. Rather, it would be a compressed empirical solution: once the model is confident about which component it is in, it may only need to track the belief state of that component. So I think a second plausible prediction is a compressed hierarchical geometry with effective dimension closer to 4 than 8.

This 4D picture is not an exact representation of the full sufficient predictive state in general. Rather, it is a hypothesis about the geometry the trained network may preferentially realize in practice, especially if component identity becomes nearly certain after a short prefix.

Prediction across context position

I expect the geometry to change systematically with context position.

At early positions, the model has seen little evidence, so uncertainty over the generating component should still be relatively high. Therefore I expect early activations to be dominated more by the component-inference part of the problem. Geometrically, this suggests more overlap between contexts from different components and stronger variation along directions associated with uncertainty over C.

At later positions, the model should become more certain which component generated the sequence. So I expect the remaining variation to be more strongly organized by the within-component Mess3 predictive state. So later in the sequence, activations should cluster more clearly by true component with finer low-dimensional variation within each cluster corresponding to the component's Mess3 belief geometry.

In short, my prediction across context position is an early-to-late transition from "which component generated this sequence?" (component posterior geometry) to "what is the current predictive state within that component?" (within-component belief geometry).

So the geometry might look like:

> *early positions: points arranged according to uncertainty over component identity; later positions: three clearer regions, one for each component; within each region: some 2D Mess3-like structure*

Prediction across layers

I also expect the geometry to sharpen across layers.

In earlier layers, I expect the representation to contain relatively coarse evidence about component identity, such as local token-frequency cues or weak regime-separation signals. These representations may be noisier and more entangled.

In later layers, I expect the geometry to become more predictive and more aligned with the latent state needed for next-token prediction. If a clean low-dimensional predictive manifold exists, it should be more visible in later residual-stream activations than in the earliest layers. So my layer-wise prediction is:

> *early layers: partial and entangled evidence about component identity; later layers: cleaner clustering by component and clearer low-dimensional within-component predictive geometry*

Geometric intuition

I think the activations should not form one unstructured cloud. Instead, they should reflect the fact that the model is doing latent regime inference.

The most literal mathematical picture is an 8-dimensional simplex-like predictive geometry over the 9 joint latent states (C, S_t). But I do not expect this to appear visually as a single simple object. More likely, I think it will appear as something hierarchical.

One possible picture is that contexts first spread out according to uncertainty over component identity, and then each component region contains its own local Mess3 belief geometry. Since each Mess3 belief state is 2-dimensional, each component contributes something triangle-like or sheet-like in activation space. Early ambiguous contexts may lie between these regions, while later more confident contexts lie closer to one component-specific region.

Another plausible picture is a more compressed geometry, where the model separates "which component am I in?" from "what is the current predictive state?" into approximately different subspaces. In that case, the geometry would look more factored, even though the data here is a mixture over components rather than the tensor-product factorization.

So, before seeing the results, I consider at least three plausible geometric outcomes (shown in Figure 3):

1. **Full predictive-simplex geometry:** activations reflect the full 8D predictive state over the 9 joint latent states (C, S_t).

2. **Compressed hierarchical geometry:** activations use roughly 4 effective dimensions, with one low-dimensional part for component identity and one low-dimensional part for the active Mess3 belief state.

3. **Union-of-manifolds geometry:** activations form three component-specific low-dimensional regions or sheets, with uncertain contexts interpolating between them rather than lying on one clean global manifold.

My formal baseline prediction is that the sufficient predictive state is 8-dimensional, corresponding to a normalized belief over the 9 latent states (C, S_t). However, my empirical prediction for the learned neural representation is: the transformer may realize this predictive structure in the form of the latter two, for example by separately encoding component uncertainty and the active component's local Mess3 belief. So I treat the 8D simplex as the clean task-level reference geometry, and the 4D/union-of-manifolds as plausible learned-representation geometries.

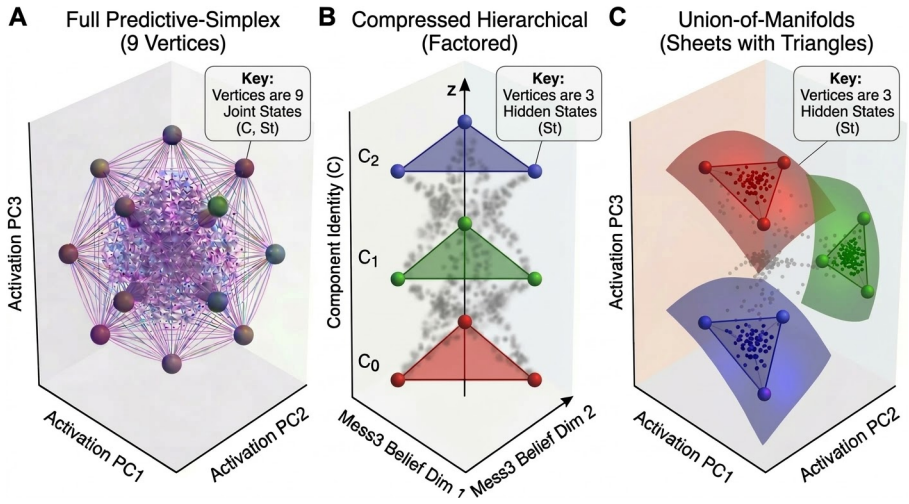

Figure 3: Three plausible geometric structures for the transformer's predictive state space. **(A) Full Predictive-Simplex:** Activations reflect the 8-dimensional predictive geometry over the 9 joint latent states (C, S_t), visualized as a projected 8-simplex along the first three principal components (PCs) of the activation space. The vertices represent the 9 distinct joint states. **(B) Compressed Hierarchical:** The representation uses roughly 4 effective dimensions, explicitly factoring the space. The vertical axis represents component identity (C), while the orthogonal horizontal plane captures the active Mess3 belief state. The vertices of the triangles correspond to the 3 hidden states (S_t) of the respective component. **(C) Union-of-Manifolds:** Activations form three component-specific sheets embedded in the global activation space (projected onto PC1, PC2, and PC3). Within each sheet, representations organize into a 2D simplex (triangle) corresponding to the component's Mess3 belief geometry, with vertices representing the 3 hidden states. In all plots, faint gray points indicate early, highly uncertain contexts interpolating between the more confident states.

3 Residual stream geometry after training

After training, I analyzed the transformer's residual stream on a held-out val set to determine what structure was present and how it related to the true belief geometries of the underlying Mess3 components.

For each val sequence prefix $x_{1:t}$, I computed the exact Bayesian latent targets induced by the data-generating process: the posterior over components $p(C \mid x_{1:t})$, and the within-component hidden-state beliefs $p(S_t \mid C = c, x_{1:t})$ for each of the three Mess3 components. I then extracted residual-stream activations from the trained transformer and compared them to these exact Bayesian targets using PCA and linear regression probes. I measured these quantities across all layers.

Low-dimensional structure across layers

I first performed PCA on the residual-stream activations at three points in the network: the input to the first block (Layer 0), the input to the second block (Layer 1), and the final residual stream before the output layer norm (Layer 2). Figure 4 shows the cumulative explained variance curves across these layers.

The main pattern is that the representation becomes much more compressed as it moves through the network. At Layer 0, the activations are fairly spread out. 7 principal components are required to explain 90% of the variance, and 10 are required to explain 95%. By Layer 1, this collapses sharply to only 2 and 3 principal components for respectively 90% and 95% variance explained. At Layer 2, the representation remains highly low-dimensional, with 2 principal components explaining 90% of the variance and 3 explaining 95%.

So although the model width is 64, the learned residual geometry at the up,per layers occupies only a very small number of effective dimensions. This already rules out the idea that the residual stream is a generic high-dimensional cloud.

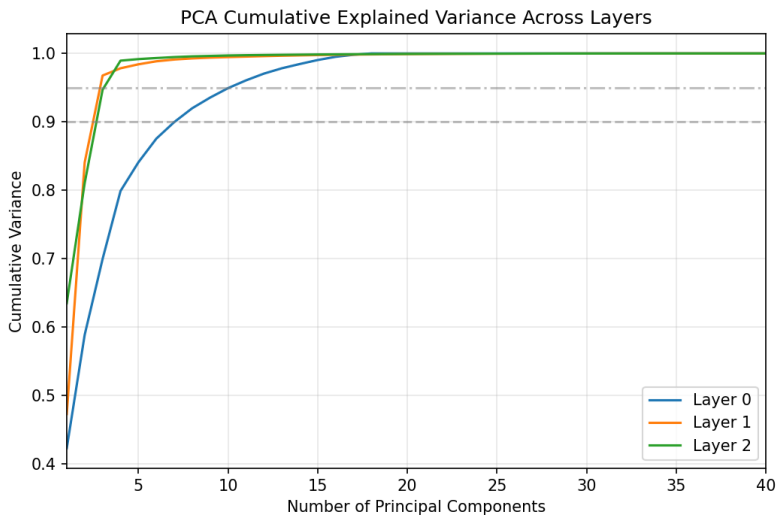

Figure 4: Cumulative explained variance of residual-stream activations across layers. Layer 0 is substantially more diffuse, while Layers 1 and 2 are strongly compressed into a low-dimensional subspace. This suggests that the transformer organizes and compresses predictive information as it processes context.

Linear decodability of latent predictive variables

To test whether this low-dimensional geometry corresponds to the true predictive latent state of the data generator, I fit linear regressions from residual-stream activations to the exact Bayesian belief variables.

Table 2 and Figure 5 show the resulting test R^2 values across layers.

Target variable	Layer 0	Layer 1	Layer 2
Posterior $p(C \mid x_{1:t})$	-0.001	0.253	0.630
Belief $p(S_t \mid C = 0, x_{1:t})$	0.789	0.960	0.957
Belief $p(S_t \mid C = 1, x_{1:t})$	0.838	0.963	0.958
Belief $p(S_t \mid C = 2, x_{1:t})$	0.815	0.888	0.850

Table 2: Linear probe performance (R^2) from residual-stream activations to exact Bayesian latent variables, measured across layers.

Two patterns stand out:

First, uncertainty over *which component generated the sequence* becomes progressively more explicit across layers. At Layer 0, the posterior over components is essentially not linearly recoverable ($R^2 \approx 0$). At Layer 1, it becomes modestly recoverable ($R^2 = 0.253$), and by Layer 2 it is clearly present ($R^2 = 0.630$). This is strong evidence that the transformer progressively infers component identity across layers.

Second, the *within-component hidden-state beliefs* are already fairly recoverable even at Layer 0, and become very strongly recoverable in deeper layers. For components $C0$ and $C1$, the Layer 1 and Layer 2 probes are near 0.96 R^2. For component $C2$, the recovery is weaker but still strong, around 0.85–0.89. This suggests that the network is not only inferring which regime it is in, but also tracking predictive state within each regime.

Although the within-component belief probes are already very strong at Layer 1, some of these R^2 values decrease slightly at Layer 2. I do not interpret this as loss of predictive information. A linear probe only measures how easily a target can be recovered by a linear map, not whether the information is present in a more compressed or more entangled form. Since the transformer is trained to optimize next-token prediction rather than to make each latent variable maximally linearly decodable, it is plausible that the final layer reorganizes the representation to better support the overall prediction objective (especially component-level regime inference) while slightly reducing linear recoverability of some individual within-component belief coordinates.

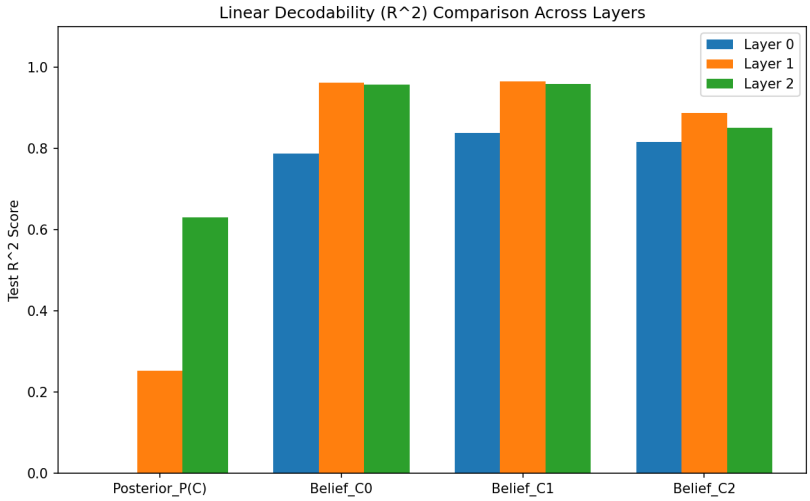

Figure 5: Linear decodability of exact Bayesian belief variables across layers. Posterior over component identity becomes much more recoverable in deeper layers, while within-component hidden-state beliefs are already present at Layer 0 and become especially strong by Layers 1 and 2.

Global structure across layers

To visualize how the learned geometry changes across the network, I projected the residual-stream activations from all three layers into their first three principal components. Figure 6 shows these 3D PCA projections colored by true generating component.

The 3D visualizations clarify the layerwise story suggested by the PCA and probe analyses.

At Layer 0, the geometry is still relatively unorganized from the perspective of component identity. The 3D projection does not show a clean compressed predictive manifold. This is consistent with the quantitative result that Layer 0 is substantially higher-dimensional than the deeper layers: it requires 8 principal components to explain 90% of the variance and 11 to explain 95%. It is also consistent with the probe result that the posterior over components is essentially not linearly decodable at Layer 0.

Because Layer 0 is substantially higher-dimensional than the deeper layers, its projection into the first three principal components should be interpreted more cautiously. Higher dimensionality can make the first three PCs explain less of the total variance, and can make the resulting 3D projection look thinner or less representative of the full geometry than comparable projections from later layers. So the Layer 0 plot is best understood as a partial view of a higher-dimensional representation, rather than as a complete picture of its structure.

By Layer 1, the geometry collapses into a highly structured low-dimensional manifold. This is the same layer where PCA shows maximal compression, with only 2-3 principal components needed to explain 90% and 95% of the variance. Although the three components are still not cleanly separated into disjoint regions, the representation is no longer diffuse. Instead, it has been compressed into a compact shared predictive geometry.

By Layer 2, the representation remains strongly low-dimensional, but the geometry broadens relative to Layer 1. This is also the layer where the posterior over components becomes much more linearly recoverable, yet the projection still does *not* split into three fully disjoint clusters So Layer 2 appears to retain the low-dimensional predictive structure learned earlier while reorganizing it into a somewhat richer final representation that makes component-level regime information more explicit.

This pattern aligns with the probe results: the posterior over components becomes much more explicit in the final layer even though visual separation remains partial. Overall, the across-layer 3D visualizations suggest that the transformer does not simply sharpen a fixed geometry monotonically. Instead, it first compresses the representation strongly by Layer 1, then reshapes it at Layer 2 into the final predictive geometry used for next-token prediction.

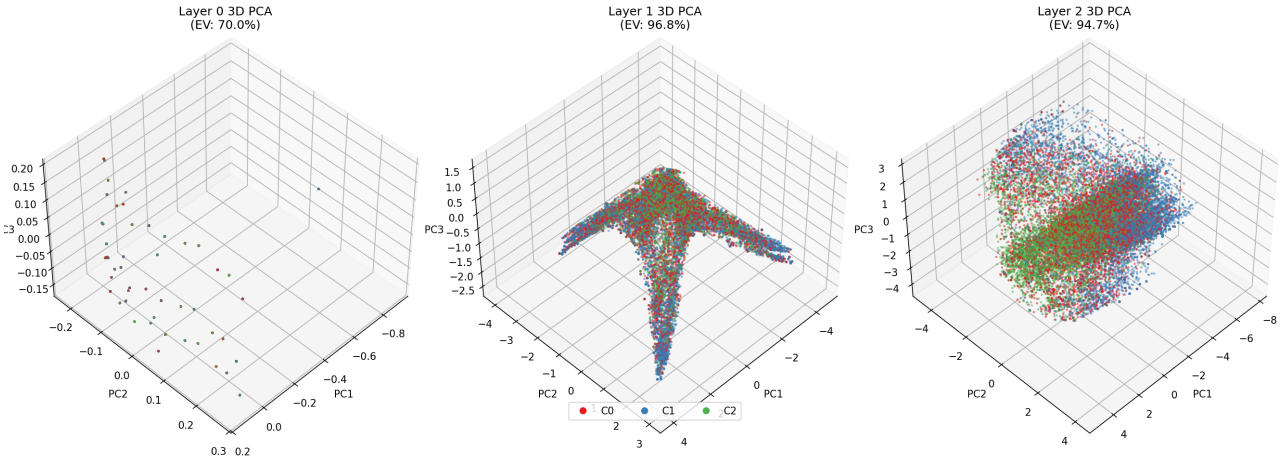

Figure 6: 3D PCA projections of residual-stream activations across layers. Each color represents one of the three generating components ($C0$, $C1$, $C2$). Layer 0 remains relatively unorganized and higher-dimensional, consistent with its larger effective dimensionality. Layer 1 becomes sharply compressed into a compact structured manifold. Layer 2 remains low-dimensional but broadens into a richer final predictive geometry with somewhat stronger component-dependent organization.

Within-component belief geometry across layers

An important question is whether the residual stream preserves meaningful *within-component* predictive geometry, rather than only tracking which component generated the sequence. To test this, I restricted to prefixes whose true generating component was $C0$, ran PCA separately at each layer, and colored the points by one coordinate of the exact Bayesian belief state, namely $p(S_t = 0 \mid C = 0, x_{1:t})$. Figure 7 shows these 3D PCA projections.

This figure shows that the within-component belief geometry also changes substantially across layers.

Interestingly, the within-component $C0$ manifold is visually similar in its overall shape to the global manifold shown in Figure 6, even though Figure 7 is computed on a strict subset of the data. I verified that the global plot uses all 34,000 prefix examples in the analysis set, while the $C0$-only plot uses only the 11,101 examples whose true generating component is $C0$. I interpret this similarity as evidence that the learned representation is not organized primarily as three completely separate component-specific clouds. Instead, the transformer appears to use a largely shared predictive geometry, within which both regime identity and component-specific belief variation are represented. This is also consistent with the probe results: within-component belief variables are very strongly decodable, whereas component identity is only moderately decodable, suggesting that belief-state variation may contribute more to the dominant geometry than regime identity alone.

At Layer 0, there is only weak visible organization in the 3D projection. This is consistent with the quantitative story: some belief information is already linearly decodable at Layer 0, but it has not yet been organized into a clean compact manifold.

At Layer 1, the within-component geometry becomes much clearer. The points lie on a sharply structured low-dimensional manifold, and the color gradient varies smoothly across that manifold. This is strong qualitative evidence that the network has organized the residual stream around the true predictive belief state of the Mess3 process.

At Layer 2, this smooth belief-dependent variation is still present, but the geometry is no longer as sharply simple as at Layer 1. Instead, it appears embedded in a somewhat broader final-layer representation. This matches the quantitative probe results: within-component belief remains highly decodable at Layer 2, but some probe R^2 values are slightly lower than at Layer 1. I do not interpret this as loss of predictive information. A linear probe only measures how easily a target can be recovered by a linear map, not whether the information is present in a more compressed or more entangled form. Since the transformer is trained to optimize next-token prediction rather than to make each latent variable maximally linearly decodable, it is plausible that the final layer reorganizes the representation to better support the overall prediction objective (especially component-level regime inference) while slightly reducing linear recoverability of some individual within-component belief

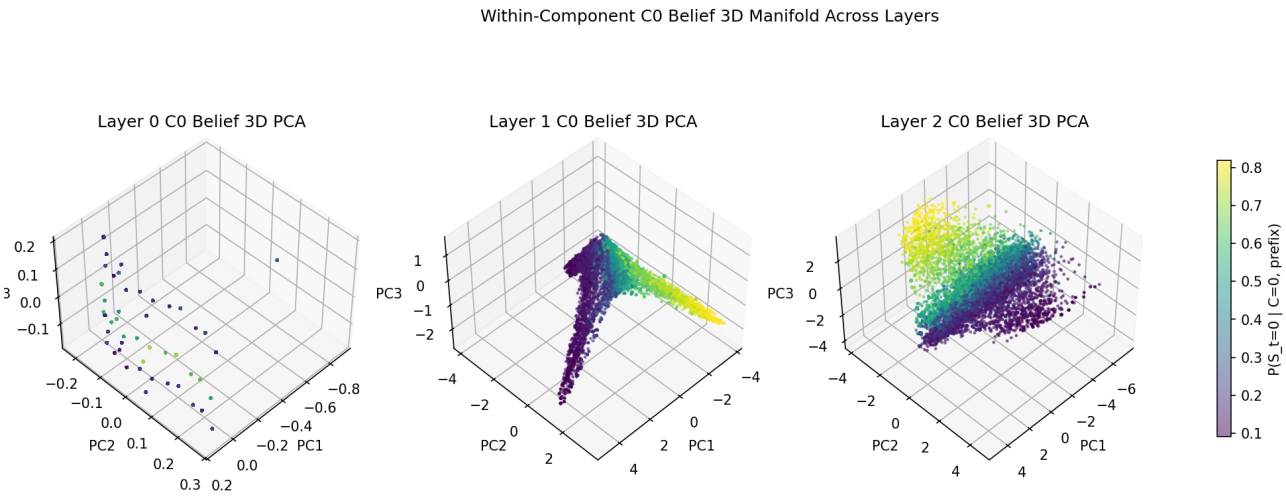

Figure 7: Within-component 3D PCA projections across layers for prefixes generated by component $C0$, colored by the Bayesian belief coordinate $p(S_t = 0 \mid C = 0, x_{1:t})$. Layer 0 shows only weak visible organization. Layer 1 shows the clearest compact belief-relevant structure. Layer 2 preserves this belief-dependent variation while embedding it into a somewhat broader final predictive geometry.

coordinates.

Taken together, these within-component 3D plots suggest that the network does not merely classify sequences by component. It also learns a genuinely belief-relevant internal geometry within each regime, and this geometry becomes most visibly structured at Layer 1 before being integrated into the final predictive representation at Layer 2.

Context position dependence

I also measured how decodability changes with context position at Layer 2. Figure 8 shows the test R^2 of linear probes as a function of prefix length.

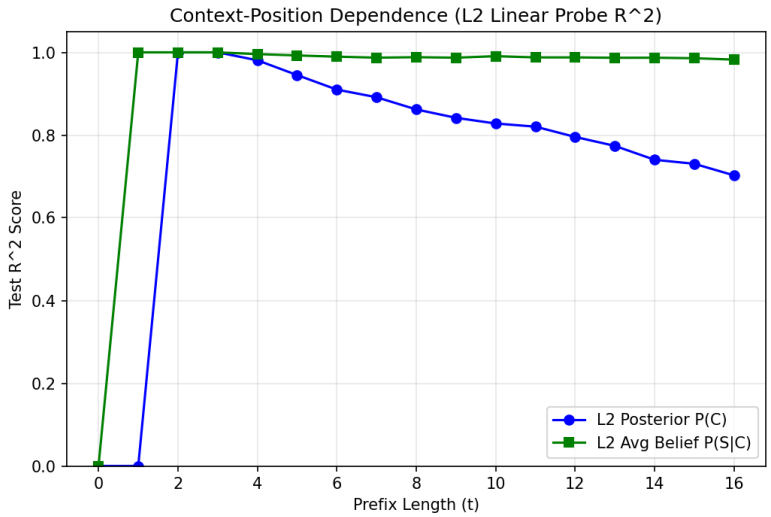

Figure 8: Context-position dependence of Layer 2 linear probe performance.

The within-component beliefs remain highly decodable across nearly all nontrivial prefix lengths. The posterior over components becomes highly decodable very quickly, but then gradually declines as prefix length

increases. I don't want to over-interpret this pattern, but maybe it is consistent with the broader layerwise story in Figure 5: deeper layers make component identity more explicit, while within-component predictive structure remains throughout and appears to dominate more of the geometry once regime identity is largely resolved. This may also help explain why some within-component belief probes are slightly stronger at Layer 1 than at Layer 2 in the figure. The final layer is optimized for next-token prediction rather than maximal linear readability of each latent variable separately, so it is plausible that once the generating regime is identified, later activations allocate more of their representational variation to the detailed predictive state within that regime.

What structure is there?

Taken together, these results suggest that the residual stream has a **compressed hierarchical geometry** rather than a generic diffuse cloud or three fully disjoint clusters.

At a coarse level, the network represents uncertainty over *which component generated the sequence*. This information is weak or absent in Layer 0, emerges in Layer 1, and becomes clearly linearly recoverable by Layer 2. At a finer level, the network also represents the *predictive belief state within each component*: the local PCA plot for component $C0$ shows smooth variation with the corresponding Mess3 belief state inside a single component region.

Each Mess3 component is a 3-state HMM, so its predictive belief state lies in a 2D simplex. The probe and visualization results indicate that this within-component belief geometry is approximately preserved in residual-stream activations. Once activations are restricted to a component-specific region, they continue to vary along directions that linearly encode the relevant Mess3 hidden-state belief.

Because the full dataset is a mixture of components, the model must represent both regime identity and within-regime predictive state. Accordingly, the learned representation is best described as a shared low-dimensional arrangement that combines (1) latent regime inference across components and (2) within-regime predictive geometry inside each component.

Relation to the belief geometries of the component processes: Each component process has its own predictive belief geometry, and the learned residual representation appears to preserve these local structures rather than collapse them. In particular, once we condition on a component, the activations vary along directions that track the corresponding within-component Mess3 belief state. This supports the view that the network is not only separating regimes at a coarse level, but also retaining belief-relevant geometry inside each regime.

This is exactly the structure expected for prediction in a non-ergodic mixture: the model must infer *which regime it is in* and track *what predictive state it is in within that regime*. Overall, the results are more consistent with the compressed/hierarchical hypothesis from part 2 than with a fully explicit 8-dimensional realization of the joint predictive simplex, even though it still preserves the relevant latent predictive information.

Summary of the main findings

The part 3 results are summarized by three main claims:

1. Low-dimensional structure exists. PCA shows that the final residual stream is strongly concentrated in a low-dimensional subspace: 3 principal components explain 90% of the variance and 4 explain 95%. So although the model width is 64, the learned representation is effectively much lower-dimensional.

2. Residual activations encode the true latent predictive variables. Linear probes recover the posterior over components moderately well and the within-component hidden-state beliefs very well. In particular, the final residual stream linearly predicts $p(C \mid x_{1:t})$ with $R^2 = 0.6298$, while the three component-specific hidden-state beliefs are recovered with $R^2 = 0.9574$, 0.9582, and 0.8504, respectively. This provides evidence that the residual stream contains the correct predictive-state information for the underlying Mess3 mixture.

3. The learned geometry looks compressed/hierarchical rather than like a fully explicit 8-dimensional realization. My formal task-level baseline in Part 2 was an 8-dimensional predictive geometry over the 9 joint latent states (C, S_t). However, the learned neural representation appears much more compressed in practice. The layerwise PCA results show that the representation becomes sharply low-dimensional

by Layer 1 and remains so at Layer 2, while the layerwise probes show that regime information becomes progressively more explicit across the network. The visualizations suggest a hierarchical organization: the model first organizes a compact belief-relevant manifold and then reshapes it into a broader final representation that better exposes component-level regime information while preserving within-component predictive structure. This interpretation is supported more strongly by the PCA and probe analyses than by clean visual clustering alone.

4 Additional analysis of my choosing

If I had time for one additional analysis, I would study how the final residual-stream geometry is mapped into logits by the output projection (the unembedding matrix). I chose this because parts 2 and 3 suggest that the transformer learns a compressed low-dimensional geometry that encodes both component identity and within-component Mess3 belief state, but linear probing only establishes observational decodability. This still leaves open the question of: which parts of that geometry are actually utilized by the model for prediction?

Concretely, I would take the final residual activations, project them into their top principal components, and then measure how moving along each principal direction changes the output logits via direct logit attribution. I would then compute the subspace alignment (for instance, using cosine similarity) to compare these logit-relevant directions against the linear probe weight vectors that recover $p(C \mid x_{1:t})$ and $p(S_t \mid C = c, x_{1:t})$.

If the same directions that best encode the Bayesian belief variables demonstrate high alignment with the unembedding vectors that most strongly affect the logits, that would suggest that the belief-relevant geometry found in part 3 is not merely decodable but functionally critical for prediction.

I think this would be highly informative because it bridges the gap between representational geometry and actual model behavior, a core challenge in interpretability. The main question would be resolved: whether the low-dimensional latent structure discovered in the activations is genuinely the causal mechanism the model uses to generate next-token predictions, rather than just a correlational byproduct of the training dynamics.

References

[1] Adam Shai, Loren Amdahl-Culleton, Casper L. Christensen, Henry R. Bigelow, Fernando E. Rosas, Alexander B. Boyd, Eric A. Alt, Kyle J. Ray, and Paul M. Riechers. *Transformers Learn Factored Representations.* arXiv:2602.02385, 2026.